Technical Handover — Knowledge Assistant V1

Oct 14, 2026 · @Hamaz Mubashar

Summary

Status: Pending Independent Acceptance. This is a technical handover and reproducible local demonstration on synthetic documents, not a production release.

- **What is demonstrably working:** the full pipeline from PDF upload to cited answer, running locally in Docker — ingestion and OCR, versioned publishing, hybrid search, role-based access isolation, prompt-injection defences, verified citations with refusal when evidence is weak, a web interface, and a reproducible evaluation and acceptance-run toolchain.
- **Evidence:** 100 automated tests passing at the frozen commit, a clean rebuild reproducing the recorded dependency fingerprint, development evaluation reports (zero access leaks in every run), and a rehearsed acceptance run. Today's re-verification is listed in the evidence index.
- **What prevents final sign-off:** (1) no independent reviewer has been nominated, so the acceptance set does not exist; (2) citation accuracy is below target on development data (92.9% vs 95%); (3) provider approval for company documents; (4) representative-document testing and deployed-environment security testing have not been done.

No model was retrained and the frozen acceptance configuration (commit `16b6215`, tag `v1-acceptance-freeze`) was not changed for this handover.

What is demonstrably working

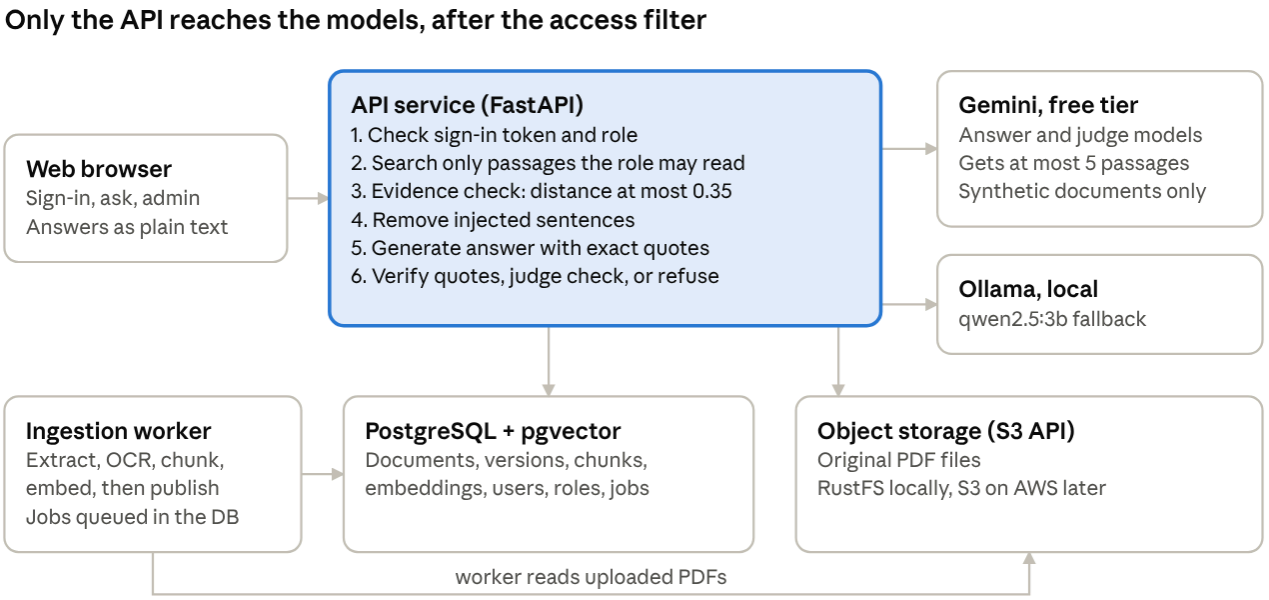
Each capability below has automated tests or a recorded evaluation behind it; "demonstrated" means shown on synthetic documents in the local environment.

Capability	Evidence
PDF upload with type and size checks; status queued → processing → published or failed with the reason; admin retry	Automated tests; web interface
Text extraction with OCR fallback for scanned pages; scanned passages flagged	Retrieval evaluation includes scanned-page questions (100% found in top 5)

Capability	Evidence
Completeness check that fails a document rather than silently losing text	Automated tests; table-row data-loss bug found and fixed 9 Oct
Versioned documents with atomic publishing; old versions never returned	0 superseded results in every retrieval evaluation
Hybrid vector and keyword search	Dev retrieval: 96.9% correct page ranked first, 100% in the top 5 (32 questions)
Role-based access isolation inside every search	0 access leaks across every retrieval and answer evaluation; restricted questions refused
Prompt-injection defences	All injection questions resisted (dev 3/3, held-out 2/2)
Verified answers: word-for-word quotes, independent judge model, refusal when evidence is weak	Frozen-configuration dev run: 92.9% correct, 0% unsupported, 7.1% over-refusal, 6/6 correct refusals
Conflicting policies: both sources cited and both values stated	Dev conflict question passes; three conflict document pairs added for acceptance
Web interface: sign-in, cited answers with Open PDF at the cited page, refusal and conflict states, document admin	Strict content-security policy, covered by tests; live demonstration (evidence index)
Reproducible build and evaluation	Locked dependencies; clean rebuild reproduces fingerprint `9b571fec3b87`; every report records code, prompt and dependency fingerprints
Sealed acceptance-run tooling	Rehearsed end to end 14 Oct: mismatch stops, match runs, second run refused

Architecture

Four services run in Docker Compose (API, worker, PostgreSQL, object storage), with the optional local model in Ollama on the host; only the API service sends text to a language model, and only after the access filter and evidence check.



System architecture · local Docker deployment

The worker processes uploads in the background, so a slow PDF never blocks questions. The embedding model runs inside both the API and the worker; no document text leaves the machine for search.

Local setup and demonstration

The system runs on one laptop with Docker; the repository's `README.md` is the authoritative copy of these steps.

1. **Prerequisites:** Docker Desktop; optionally Ollama with `ollama pull qwen2.5:3b` for the local model.
2. **Get the code:** clone `hamazmubashar/knowledge-assistant` and check out tag `v1-handover`. Its application code, Docker configuration and dependency lock are identical to `v1-acceptance-freeze`; it adds the README, `.env.example` and evidence.
3. **Configure secrets** in a `.env` file that is never committed: `JWT_SECRET` (at least 32 characters), storage access keys, and `GEMINI_API_KEY` for the free tier (synthetic documents only).
4. **Build and start:** `docker compose up -d --build`. The image installs exact package versions from `requirements.lock` on a pinned base image and builds the embedding model in.
5. **Create the database schema:** `docker compose exec api alembic upgrade head`.
6. **Create demo users** with the user-creation script (passwords are prompted, never passed on the command line).

7. **Verify:** `docker compose exec api python -m pytest -q` (100 tests) and `./scripts/eval.sh dev` (retrieval evaluation on the synthetic corpus).
8. **Demonstrate** at `http://localhost:8000`: sign in as an HR admin and ask a policy question, open the cited PDF page, sign in as a support user and confirm HR content is refused, and upload a synthetic PDF through the Documents tab.

Deployment approach and operating-cost assumptions

Nothing has been deployed beyond the local environment, and no paid resources exist; the plan below is a design, not a deployed system.

Target deployment (previously approved design, not built):

Local component	Planned AWS equivalent
API and worker containers	Amazon ECS on Fargate, one service each
PostgreSQL + pgvector	Managed PostgreSQL with the pgvector extension
RustFS object storage	Amazon S3, private bucket, encrypted
`.env` secrets	A managed secrets store, injected at runtime
Gemini free tier	Gemini on Vertex AI, subject to the provider approval request

The containers already use the same images, environment-variable configuration and S3-compatible storage interface, so the move is configuration and infrastructure work rather than code changes — apart from the Vertex AI connection.

Operating-cost assumptions: the provisional cost model (`Enterprise_Knowledge_Assistant_Cost_Model.xlsx`) remains the reference. It was accepted as provisional only; official AWS and Vertex AI pricing must be verified before any budget is approved. Model calls are expected to dominate variable cost and scale with questions asked; the evidence check avoids model calls entirely for questions with no relevant documents. No spending occurs without written approval.

Security considerations

The controls below are implemented and tested locally; none has been tested in a deployed environment or by an independent security reviewer.

Area	In place	Still needed before production
Authentication	Argon2 password hashing; signed tokens with a 32-character minimum secret; token kept in page memory only	Secure session cookies, admin password reset, single sign-on if required
Authorisation	Default-deny role permissions applied inside every search query; restricted documents answer "not found"	Review of the permission matrix by HR and IT
Prompt injection	Suspicious sentences removed before generation and filtered from answers; flagged chunks visible to admins	Testing against real documents and new attack patterns
Web interface	Strict content-security policy; model text never rendered as HTML; no external scripts	Penetration test of the deployed site
Data leaving the system	Only up to 5 permitted passages per question, synthetic documents only	Provider approval (Vertex AI request), contract terms, region choice
Secrets	Kept in an uncommitted `.env`; provider key only in the API service	Managed secrets store, key rotation
Supply chain	Locked package versions and pinned base image	Checksums for downloaded model files; dependency vulnerability scanning

Known limitations

All limitations from the freeze record and readiness package still apply; none was changed for this handover.

- Citation accuracy below target on development data (92.9% vs 95%), caused by occasional off-topic extra sentences.
- One remaining development over-refusal above the 0.35 cut-off.
- Development results come from 21 synthetic questions and are diagnostic, not acceptance evidence.

- Gemini model names are not version-pinned; downloaded model files are not checksum-verified.
- Free-tier rate limits make answers take minutes; latency has not been measured on production infrastructure.
- Keyword-based scoring can misjudge correctly worded answers (errata E1).
- The run-once check runs after the evaluation database is rebuilt rather than before (harmless; to be corrected after acceptance).
- Synthetic documents only: real layouts, scans and tables are untested.
- On a first install the worker starts before the database schema exists and exits; it has no restart policy. The README adds `docker compose restart worker` after the migration. A permanent fix (restart policy or retry on startup) changes frozen files and is scheduled after acceptance. Found by the clean-room reproduction on 15 Oct.

Dependencies blocking sign-off

Final sign-off is blocked by one missing input and several approvals, none of which development work can resolve.

Dependency	Owner	Blocks	Status
Independent reviewer to write and seal the acceptance set	Alex Morgan to nominate	Acceptance decision	No reviewer nominated
Sealed acceptance set and separate fingerprint	Independent reviewer	Acceptance run	Not started
Provider approval for company documents (Vertex AI)	IT security, legal	Real-document testing, production	Request submitted 14 Oct
Contract terms, region and excluded document types	Legal, HR	Provider approval	Open
Budget for cloud and model costs	Finance	Any deployment	Cost model provisional only
Representative-document testing	Development, after approval	Production readiness	Not started
Deployed-environment security testing	IT security	Production readiness	Not started

Until the first two rows are complete, the project remains **Pending Independent Acceptance**.

Evidence index

Today's re-verification (14 October, 15:12 PKT) confirms the frozen code is unchanged, all tests pass and retrieval results are identical to the freeze.

Check (14 Oct)	Result
Code at handover	`fb7fcd4`, seven commits after `v1-acceptance-freeze` (permissions, evidence, reports, README and .env.example only); this document is stored in the repository and tagged v1-handover in a following documentation-only commit
Frozen code unchanged	`git diff` against the tag for `app/`, `docker-compose.yml`, `Dockerfile`, `requirements.lock`: no differences
Services	API, worker, PostgreSQL and object storage all running
Automated tests	100 passed
Running settings	Evidence cut-off 0.35, reranker off, judge `gemini-3.5-flash-lite`
Retrieval evaluation (report `dev-20261014-101216.md`)	96.9% correct page first, 100% in top 5, 0 leaks on 5 restricted questions, 0 superseded results
Demo: support user asks an HR question	Refused: "No answer found in your documents"
Demo: Open PDF from a citation	Opened the cited Annual Leave Policy page
Demo: HR admin asks about annual leave (web UI)	Answered "20 days", citing EVAL HR Annual Leave Policy v1 page 1 with a word-for-word quote; Open PDF landed on page 1 (screenshots)

Check (14 Oct)	Result
Demo: upload a synthetic PDF, then ask about it (HTTP API as HR admin)	1-page synthetic visitor-badge PDF published to General; /answer replied "valid for 8 hours", citing it at page 1 (distance 0.19, 0 sentences dropped). Answered by qwen2.5:3b, judged by gemini-3.5-flash-lite
Test warning	1 Starlette deprecation warning, raised when the FastAPI test client is imported; tests only, left unchanged because fixing it would alter the frozen dependency lock
Database schema	alembic current = 17ff2353d473 (head). Migrations are not run on startup; the README covers this step
Evaluation tooling isolated from live data	app.scripts.ask refused to run against the live database ("DATABASE_URL must point to the knowledge_eval database")
Demo database cleanup	Removed 10 documents and their stored PDFs: 8 leftovers from smoke tests, one stuck-upload test, and one non-corpus project report ("Local Inference Demo — Results and Recommendation") that had been uploaded to the General group by mistake. While it was there, any question that retrieved its passages would have sent them to the Gemini judge. Deleted in one transaction; 0 orphaned versions; 21 synthetic documents remain
Clean-room reproduction (15 Oct)	Fresh clone of fb7fcd4, .env built only from .env.example (no keys missing), no-cache Docker rebuild, migration of an empty database to head, worker restart per README, 100 tests passed, retrieval evaluation 96.9% / 100% / 0 leaks / 0 superseded, frozen-path diff 0, requirements.lock sha256 a1c13ccf…ecb041. The first attempt found the worker start-up issue now in Known limitations. Log: docs/handover/clean-room-20261015.log

Demo model differs from the evaluated model. The web application currently answers with the local `qwen2.5:3b` model; the frozen evaluation configuration uses Gemini 3.8 Flash. The demo therefore shows the working pipeline, access control and verification, but its answer quality is lower than the evaluated configuration (qwen scored 50% correct on development data). Switching the demo to Gemini 3.8 Flash is a settings change, not a code change, but each answer takes minutes on the free tier.

Records: Reranker evaluation · Freeze record · Readiness package · Provider approval request · evaluation reports in `eval/reports/` · dry run in `eval/dryrun/` · errata in `eval/ERRATA.md`.